Address Resolution Protocol (ARP)

1) Why do we need both MAC Address & IP Address?

Every device connected to a network has two different addresses:

IP address	- Identifies a device across different network	Network Layer (Layer 3)
MAC address	- Identifies a device within the local network	Data link Layer (Layer 2)

* Both addresses are necessary but they solve different problems.

2) Why isn't an IP address alone enough?

When data reaches your LAN, devices communicate using MAC addresses, not IP addresses.

3) why Ethernet and Wi-Fi need MAC address?

* Many devices are connected to the same switch.

* suppose PC1 wants to send data

which device should receive it?

* without MAC address the switch cannot know.

Therefore,

Every ethernet device must have a unique MAC address.

4) Switch keeps MAC address table

But how does a computer know another computer's MAC address?

suppose

Laptop
IP = 192.168.1.2
↓
needs to communicate with server

IP = 192.168.1.3

The laptop only knows destination IP
192.168.1.3

But Ethernet requires a destination MAC address.

How can it find it?

=> ARP

5) what is ARP?

Its Job is, convert an IP Address into a MAC address.

IP address ⟶ ARP ⟶ MAC address

6) ARP cache

* Every computer stores recently discovered MAC addresses.

* This storage is called ARP cache.

* This avoids asking ARP repeatedly.

7) How it converts?

The Request : when a device wants to send data to a new device on the same local network, it broadcasts a query.

"who has this IP address"

ARP request is broadcast

Destination MAC : FF:FF:FF:FF:FF:FF

The Response : The device with that matching IP address replies directly with its specific MAC address.

* ARP reply is <u>UNICAST</u>

8) windows commands :

(i) see ARP table

arp -a

(ii) delete ARP table

arp -d

9) The Ethernet cable carries Ethernet Frames , not raw IP packets.

* Ethernet Frame look like this.

Destination MAC
Source MAC
Type
IP packet

* The switch only looks at destination MAC.

* It does not open the IP packet.

* It does not check destination IP

So the computer / device that needs to send data, first learns MAC address of the destination by raising a query.

Correct sentence:-

The computer learns the MAC because Ethernet switches forward frames using MAC addresses, not IP addresses. without destination MAC, the switch cannot deliver the Frame to the correct device.

If the switch ignores IP, then who uses the IP?

* The sender :- uses the destination IP to determine where it wants to send the packet.

* Router : use IP addresses to forward packets between different networks.

* The destination computer checks the destination IP after receiving Ethernet frame.

So :-

* Switch → MAC address
* Router → IP addresses
* Host (computer) → uses both